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Full Length Article

Investigating the impact of biochar on methane gas emissions and its effect on enteric fermentation

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ABSTRACT

This study aims to evaluate the impact of biochar on methane gas emissions and to investigate its effect on enteric fermentation. Methane gas emissions from Naeemi ewes' ruminants were measured, with animals divided into two groups, each containing 14 ewes. The treatment group was fed a combination of biochar and a basal ration, while the control group received the basal ration alone. The results demonstrate that the initial gas emission for both animal groups at day 0 was 1082 ppm, and the inclusion of biochar in the diet significantly ($P < 0.05$) reduced CH_4 gas emissions by 65.58 %–78.39%. Furthermore, feeding Naeemi sheep with biochar and the basal ration led to a significant ($P < 0.05$) increase in body weight (63.66 kg) and growth rate (length: 79.3 cm, waist: 81.75 cm, and height: 99.76 cm) compared to the control group. Additionally, it raised the average body condition score (BCS) to 3.56, in contrast to the control group's score of 2.83, primarily due to the positive impact of biochar on fat gain. The study revealed that biochar enhances nutrient utilization, improves feed efficiency, and significantly reduces methane gas emissions in livestock. When incorporated into the sheep's diet, biochar can modify digestive processes and promote better nutrient absorption.

1. Introduction

It has been documented that biochar is a non-toxic substance, making it suitable for incorporation into animal diets (Sakadevan and Nguyen, 2017; Das et al., 2020; Llonch et al., 2016; Tang et al., 2019; Joseph et al., 2018). Research studies have shown that the oral consumption of biochar by various species can enhance nutrient utilization, adsorb harmful substances, and generally improve animal well-being (Joseph et al., 2018; O'Toole et al., 2016; Kammann et al., 2015). One study was conducted on a poultry farm in Kuwait, using biochar as a feed additive to examine its effect on poultry production and performance. This research focused on the different biomaterials used in biochar production and various other products generated through processes such as pyrolysis, thermochemical conversion, hydrothermal processing, and microwave techniques. Additionally, the analysis explored synthetic processes for producing biochar and its numerous applications in the poultry industry (Al-Khalaf and Nasser, 2023). No prior studies on the use of biochar in sheep in Kuwait or GCC countries were found. Other studies have been conducted in various Arab countries, including Qatar, Iraq, and Saudi Arabia. However, in these studies, biochar production from different biomass sources was primarily aimed at improving and increasing crop production in the agricultural sector, rather than being used as livestock feed (Elkhalifa et al., 2019; Parthasarathy et al., 2023; Diatta et al., 2020).

On the other hand, many studies were conducted in Kuwait investigating greenhouse gas emissions but from different sources other than from the livestock sector between 2004 and 2016 in four major urban cities of Kuwait. The results showed a clear increase in greenhouse gas emissions, with higher emissions in summer compared to winter. This increase is likely due to the high consumption rate of fossil fuels for cooling purposes and increased traffic activities during the summer months (Harbi et al., 2020). In addition, another study evaluated the average carbon dioxide emissions resulting from fossil fuel combustion over the period from 2009 to 2019. The study focused on Iraq, Saudi Arabia, and Kuwait, as these countries have the largest fossil fuel production and storage. The results showed that Saudi Arabia had the highest annual emission rate, reaching 645.4 million tons in 2015. Kuwait had the highest emissions per capita, with 28.1 tons annually, followed by Saudi Arabia. The share of global carbon dioxide emissions was 1.54% for Saudi Arabia, 0.47% for Iraq, and 0.29% for Kuwait (Al-Naffakh et al., 2021). To date, no study has been conducted to measure greenhouse gases emitted from the livestock sector in Kuwait or

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the broader Arab region.

Therefore, the present study aims to manage methane emissions from the livestock sector by incorporating sterile date palm biochar into animal feed. The underlying principle is that during enteric fermentation, the presence of biochar causes H^+ ions to react with nitrogen to form NH_4 instead of methane, which would typically result from its reaction with carbon dioxide (CO_2) (Osman et al., 2022). The ammonium (NH_4) is then adsorbed onto the biochar due to its high cation exchange capacity, facilitated by functional groups carrying negative charges. Consequently, supplementing livestock diets with biochar is anticipated to reduce methane emissions (Luna et al., 2018). Additionally, the inclusion of biochar is expected to improve sheep health and benefit local food production systems by reducing methane emissions (IPCC, 2013). The biochar used in this study was produced via pyrolysis, a process that involves subjecting agricultural waste to high temperatures (300–600 °C) in the absence of oxygen, as described by (Downie et al., 2012) and (Tapio et al., 2017). We hypothesize that using biochar in animal rations could reduce methane emissions and improve animal health.

2. Materials and methods

2.1. Animals

Two field experiments were conducted at two locations: the livestock research area at the Kuwait Institute for Scientific Research (KISR) farm and the Public Authority of Agriculture Affairs and Fish Resources (PAAFR). Each location housed two groups of animals, with each group consisting of 14 adult ewes approximately 1-year-old. The ewes were individually housed and received two daily feedings, one in the morning and one in the afternoon, with a total ration of 1 kg per ewe per day. One group received a basal ration, while the other group received a ration supplemented with 1% biochar per animal per day. Despite these treatments, no significant differences were observed in the collected data from both farms. Therefore, the data presented in this manuscript were obtained from the KISR farm. The research involving Naeemi sheep strictly adhered to ethical guidelines and obtained approval from the Project Review Committee at the Kuwait Institute for Scientific Research under project code FA187C. Due to the absence of a dedicated ethical committee at our institute, traditional ethical approval procedures were not pursued or granted. Nevertheless, we ensured the ethical treatment and well-being of the animals throughout the study by adhering to ethical principles.

2.2. Preparation of basal ration

The basal ration used in the present study is composed of 70% concentrate and 30% roughages. The content of the basal ration is shown in Table 1.

2.3. Proximate analysis of basal ration

Feed samples were collected for quantitative analysis to determine the percentages of the following components: moisture, crude protein, crude fat, fiber, and ash (Table 2). The ration used for ewes had a concentrate-to-roughages ratio of 70:30, which was found to be the best feeding regime for the complete production cycle of ewes in Kuwait's feedlot system, regardless of the season (Abbas et al., 2015).

2.4. Production of biochar from date palm fronds

Date palm fronds were collected from the KISR premises and chopped into suitable sizes to feed into the biochar production machine. The date palm biochar was produced through the pyrolysis process at high temperatures (550 °C) in a depleted oxygen environment. The prepared biochar was then ground to pass through a 2 mm sieve for uniformity (Pereira et al., 2014).

Table 1
Ingredients of rations on dry basis, vitamin, and mineral composition.

Ingredients	(70 Concentrate:30 Roughages) (Used for ewes and rams)
Concentrates	
Barley	40.5
Wheat Bran	10.0
Corn	10.0
Soybean Meal	6.5
Vitamins and minerals (premix[a])	1.0
Limestone	1.0
Salt	1.0
Total A	**70.0**
Roughages	
Alfalfa hay	15.0
Wheat straw	15.0
Total B	**30.0**
Total A + B	**100**

[a] Phosphorus (5.00 %), calcium (18.00 %), sodium (5.00 %), magnesium (5.00 %), manganese (500 mg/kg, manganese oxide), cobalt (100 mg/kg as cobaltous sulfate), zinc (2000 mg/kg, as zinc oxide), iodine, 125 mg/kg (as calcium iodate), selenium (10 mg/kg, as sodium selenite), vitamin A (400000 IU/kg), vitamin D3 (100000 IU/kg), vitamin E (Alpha–Tocopherol, 400 IU/kg).

Table 2
Composition of basal ration.

Component (%)	Composition of ration (70 Concentrate: 30 Roughages) (For ewes)
Moisture	8.03 ± 0.13
DM	90.98 ± 0.1
Ash[a]	5.48 ± 0.18
CP[a]	13.32 ± 0.29
EE[a]	1.56 ± 0.84
NDF	20.11 ± 0.56
ADF	4.95 ± 0.28
Ca[a]	0.37 ± 0.051
P[a]	1.26 ± 0.04

[a] DM: % of dry matter; CP: crude protein; EE: ether extract; ADF: acid detergent fiber; NDF: neutral detergent fiber; Ca: calcium; P: phosphorous.

2.5. Preparation of new ration

The pelleted animal feed was ground to pass through a 2 mm sieve to transform it into a uniform-sized powder for mixing with biochar. The biochar was uniformly mixed with the ground feed at a rate of 1% (10 g of biochar per 990 g of animal feed) (Silivong and Preston, 2016).

2.6. Assessment of methane gas emissions

The study was conducted at the research facilities of KISR and the PAAFR research farm. Methane gas emissions were quantified using respiration chambers, following the design and methodology outlined in Pinares-Patiño et al. (2012). Fourteen female fat-tailed Naeemi sheep, aged between 11 and 12 months, were measured within the respiration chambers. The methodology for methane gas measurement described by Pinares-Patiño et al. (2011) was employed. The animals underwent a 21-day acclimatization period during which they were introduced to a standard basal ration. Over 63 days, methane gas emissions were monitored. The animals received their feed in two equal portions each day, at 5 a.m. and 11 a.m. They were randomly divided into two groups, each consisting of 14 animals.

3. Results and discussion

3.1. Body weight and growth rate

The results demonstrated that feeding Naeemi sheep with a basal ration + biochar significantly ($P < 0.003$) enhanced body weight and growth rate (length, height, waist) compared to the control group. The average body weight of animals fed with a basal ration + biochar reached 63.66 kg, whereas the average body weight of animals in the control group was 58.91 kg (Table 3). Furthermore, feeding the animals with a basal ration + biochar significantly increased their growth rate (length, waist, height) as follows: 79.3 cm, 81.75 cm, and 99.76 cm, respectively, compared to the control groups (Table 3).

3.2. Body condition score (BCS)

The animals that were fed with a basal ration + biochar exhibited a significantly higher average BCS, reaching 3.56, in comparison to the control group which had a score of 2.83. Numerous research studies have consistently demonstrated the positive impact of supplementing sheep feed with biochar on fat gain. Biochar, a type of charcoal derived from biomass, has been found to improve nutrient utilization and enhance feed efficiency in livestock. When incorporated into the sheep's diet, biochar can modify the digestive processes and promote the absorption of nutrients, including fats. This increased availability of nutrients contributes to greater fat deposition in the animals (Table 3).

3.3. Total feed consumption

The primary objective of this study is to develop a new sheep ration aimed at reducing methane gas emissions during the digestion process of ruminants. Table 4 presents the total feed consumption ratio (FCR) for the two groups of animals. FCR, or feed conversion rate, measures the efficiency with which livestock convert animal feed into desired outputs such as meat. In the control group, the FCR was 15.11 ± 18.52 (Table 4). Conversely, in the treatment group, there was a significant increase to 17.13 ± 0.79. The results indicate that supplementing the ration of Naeemi sheep with biochar significantly increased body weight, growth rates (length, height, waist), and body condition scores compared to the control group. These findings are consistent with those reported in previous studies (Pinares-Patiño et al., 2012).

The observed improvements can be attributed to the effects of biochar, which enhance feed efficiency by improving nutrient digestibility and absorption in animals. Biochar also functions as a binding agent, reducing nutrient loss through excretion and thereby enhancing overall feed efficiency. Additionally, biochar promotes livestock growth and weight gain, as suggested by prior research (Pinares-Patiño et al., 2011). The enhanced availability and absorption of nutrients contribute to increased body weight and overall productivity. Moreover, the incorporation of biochar into animal rations has been shown to improve gut health, as evidenced by the absence of mortality or morbidity recorded during this study. Biochar may act as a prebiotic, creating a favorable environment for beneficial bacteria in the gut, thereby enhancing the immune system and potentially reducing the risk of certain diseases (Pinares-Patiño et al., 2011; Hansen et al., 2012). Furthermore, biochar can bind to toxins, heavy metals, and mycotoxins in the digestive tract, reducing their absorption and improving animal health. However, it is essential to note that excessive biochar amounts or improper formulations could potentially interfere with mineral absorption, leading to nutrient imbalances (Leng et al., 2012a, 2012b; Moss et al., 2000).

Numerous studies have reported similar findings, involving different animal species and biochar from various sources. For example, a study by Villalba et al. (2002) fed goats a ration supplemented with bamboo biochar at a rate of 1g per 1 kg of body weight. The findings were consistent with our current study, showing a significant increase in average daily body weight gain, with an average of 53 g compared to 44 g in the control group. This could be attributed to biochar facilitating the digestion of these leaves by absorbing their tannins, thereby increasing crude protein intake. Additionally, previous studies investigated feeding

Table 4
The effect of feeding Naeemi sheep with basal ration and ration supplemented with biochar on total feed consumption (kg/2 weeks).

Time[a] (Every 2 weeks)	Treatments	
	Control (Basal ration only)	Basal ration + Biochar
0	14.34 ± 1.41	14.24 ± 1.40
2	14.26 ± 1.61	16.37 ± 1.24
4	14.99 ± 0.83	17.97 ± 0.39
6	15.41 ± 0.46	18.06 ± 0.13
8	16.57 ± 0.29	18.99 ± 0.23
Overall averages	**15.11 ± 18.52[a]**	**17.13 ± 3.79[b]**

[a] Data were collected every two weeks, (n = 14), values are the means ± SD, all animals are females.

Table 3
The effect of feeding Naeemi sheep with basal ration and ration supplemented with biochar on body weight (kg), body length (cm), body height (cm), body waist (cm), and body condition score.

		Time[a] (every 2 weeks)								
		0	2	4	6	8	10	12	14	Average
Control	**Length**	73.5 ± 3.7[b]	73.6 ± 4.6[a]	74.4 ± 4.3[a]	74.8 ± 3.0[a]	75.6 ± 4.8[a]	76.4 ± 4.5[a]	76.7 ± 4.3[a]	76.8 ± 3.6[a]	**75.21[a]**
	Height	93.5 ± 6.2[a]	93.9 ± 6.4[a]	94.1 ± 4.1[a]	94.2 ± 5.6[a]	94.3 ± 2.7[a]	94.4 ± 3.4[a]	94.5 ± 3.5[a]	94.8 ± 3.5[a]	**94.21[a]**
	Waist	76.3 ± 8.9[a]	77.5 ± 4.6[a]	78.2 ± 3.8[a]	78.2 ± 5.5[a]	78.7 ± 4.1[a]	78.8 ± 3.9[a]	78.9 ± 2.2[a]	78.9 ± 2.4[a]	**78.17[a]**
	Body Weight	58.4 ± 7.3[a]	58.4 ± 8.5[a]	58.5 ± 10.1[a]	58.7 ± 8.4[a]	58.9 ± 7.6[a]	59.1 ± 9.5[a]	59.6 ± 8.4[a]	59.7 ± 6.5[a]	**58.91a**
	BCS	2.7 ± 0.6[a]	2.8 ± 0.6[a]	2.9 ± 0.6[a]	2.9 ± 0.6[a]	2.9 ± 0.6[a]	2.9 ± 0.7[a]	2.9 ± 0.7[a]	2.9 ± 0.7a	**2.83a**
Basal ration + biocar	**Length**	73.2 ± 2.5[a]	75.3 ± 2.9[b]	77.3 ± 3.6[b]	79.1 ± 2.8[b]	80.1 ± 2.9[b]	82.9 ± 2.9[b]	83.2 ± 2.5[b]	83.5 ± 2.6[b]	**79.3[b]**
	Height	93.0 ± 5.6[a]	94.8 ± 3.9[a]	97.1 ± 2.7[b]	100.3 ± 4.2[b]	102.5 ± 2.3[b]	102.7 ± 3.0[b]	102.8 ± 2.7[b]	104.9 ± 6.4[b]	**99.76[b]**
	Waist	77.0 ± 3.3[a]	79.1 ± 2.8[b]	81.9 ± 4.0[b]	82.8 ± 2.5[b]	83.1 ± 1.5[b]	83.1 ± 3.0[b]	83.2 ± 2.9[b]	83.7 ± 2.8[b]	**81.75[b]**
	Body Weight	58.1 ± 6.3[a]	60.1 ± 6.8[a]	62.7 ± 7.8[b]	63.4 ± 8.2[b]	64.9 ± 7.1[b]	65.9 ± 4.4[b]	66.6 ± 5.5[b]	67.6 ± 6.7[b]	**63.66[b]**
	BCS	2.9 ± 0.4[a]	3.0 ± 0.5[a]	3.5 ± 0.7[b]	3.7 ± 0.7[b]	3.8 ± 0.6[b]	3.9 ± 0.7[b]	3.9 ± 0.6[b]	3.9 ± 0.6[b]	**3.56[b]**

[a] *: Data were collected every two weeks, (n = 14), values are the means ± SD, all animals are females.

goats a 1% biochar-supplemented ration (Tamayao et al., 2021; Man et al., 2020; Herbertson, 2018). The biochars used in these studies were derived from three different spruce sources and treated post-pyrolysis with zinc chloride, a hydrochloric acid/nitric acid mixture, or sulfuric acid, each included at 20 g/kg of diet DM. The results demonstrated a 27% increase in body weight over a 100-day trial period. Another study investigated the effect of corncob biochar supplementation on the growth performance and carcass characteristics of sheep. The findings revealed that dry matter intake, crude protein intake, dry matter digestibility, and acid detergent fiber digestibility were significantly higher in the biochar-supplemented group compared to the control group. Additionally, corncob biochar supplementation significantly improved the daily weight gain and feed conversion ratio of the rams (Keba et al., 2023).

Furthermore, Lind et al. (2024) reported similar results to the present study, as the supplementation of lamb ration with walnut shell biochar (1%) (WSB), pistachio by-product biochar (1%) (PBB), and chicken manure biochar (1.5%) (CMB), significantly ($P < 0.01$) improved the average daily gain and feed conversion ratio compared to the control diet. The digestibility of dry matter, organic matter, crude protein, and neutral detergent fiber increased significantly ($P < 0.01$) with the addition of WSB compared to the control. Ruminal ammonia levels were higher in lambs fed either WSB or PBB compared to the control ($P < 0.05$), which proves that biochar can be incorporated into fattening lamb diets as a low-cost feed additive to manipulate ruminal fermentation and improve feed efficiency, consequently enhancing animal performance. In addition, a study by Besharati et al. (2020) on broiler chickens revealed that including biochar in their diet enhanced feed conversion efficiency and reduced feed intake, leading to a more favorable feed consumption ratio. Similarly, another study involving growing rabbits found that incorporating biochar into their diet improved feed efficiency and lowered feed intake, also resulting in a better feed consumption ratio (Nawaz et al., 2020).

Animals fed a biochar-supplemented ration showed a significantly higher BCS compared to the control group. These findings align with the published results by McAvoy et al. (2020) which demonstrated that supplementing sheep feed with biochar positively impacts fat gain by improving nutrient utilization and enhancing feed efficiency in livestock. When included in the sheep's diet, biochar modifies digestive processes and promotes the absorption of nutrients, including fats. This increased nutrient availability contributes to greater fat deposition in the animals. A study conducted by Schmidt et al. (2019) observed a significant increase in fat accumulation in sheep fed a biochar-supplemented diet compared to a control group. Similarly, McAvoy et al. (2020) reported higher fat content in the carcasses of sheep that received biochar-supplemented feed. The specific mechanisms by which biochar influences fat gain in sheep are still being investigated. However, it is believed that biochar enhances nutrient absorption, modifies gut microbial populations, and improves overall digestion, all of which contribute to the observed increase in fat deposition. In summary, incorporating biochar into sheep feed has shown a positive impact on fat gain. Additionally, feeding lambs with varying levels of biochar supplementation resulted in higher fat deposition compared to the control group.

However, our results do not agree with some previous studies. In a previous trial, goats were given a feed additive of 1.5% and 3% activated coconut biochar and there was no significant improvement in feed intake (Al-Kindi et al., 2016). Additionally, a study on weaned piglets found that dietary supplementation with biochar did not influence feed intake or feed conversion efficiency (Sivilai and Prestonl, 2019). In a study by Zhang et al. (2017) on pig feed supplementation with biochar, the results indicated no significant impact on feed intake or feed conversion ratio.

Therefore, the impact of incorporating biochar into animal feed on the feed consumption ratio can vary depending on several factors, including the type of animals, feed composition, biochar dosage, and specific experimental conditions. Studies have indicated both positive and negative effects on the feed consumption ratio when biochar is included in animal feed. Some research suggests that adding biochar to animal feed can improve the feed consumption ratio, while other studies do not find significant benefits.

3.4. Reduction of methane emissions

The experiment spanned 63 days, during which weekly measurements of methane gas emissions were taken from 28 Naeemi ewes. Table 5 presents the weekly averages of the collected data. Initially, on Day 0, methane gas emissions measured 1082 ppm for both groups. The results indicated a statistically significant reduction ($P < 0.05$) in methane gas emissions in the treatment group that received a biochar-supplemented ration. This reduction ranged from 66.98% on day 7–77.63% on day 63 compared to the control group (Table 5). In contrast, the control group maintained average methane gas emissions of 1082 ppm on day 0, with readings fluctuating between 1020 and 1096 ppm throughout the experimental period, showing no statistically significant difference ($P > 0.05$) in methane gas emissions over time (Table 5). These findings are consistent with previous studies, which suggest that biochar supplementation in animal feed holds promise as a method to mitigate methane emissions from livestock (Elghandour et al., 2019). Methane, a potent greenhouse gas released during the digestive process in ruminant animals, may be influenced by biochar, which can alter gut microbial populations and inhibit methane production. However, the impact of biochar on digestibility and methane emissions can vary depending on factors like biochar type and quality, feed composition, and management practices. Previous reports also indicate that including biochar in lamb diets can enhance the digestibility of dry matter, organic matter, crude protein, and neutral detergent fiber (Mirheidari et al., 2020).

The results of the current study align with numerous prior studies demonstrating the beneficial effects of biochar supplementation on feed efficiency. For example, research has illustrated that incorporating biochar into the diet of dairy cows improved feed efficiency and reduced methane emissions (Connor, 2015). Similarly, another study found that adding biochar to the diet of beef cattle enhanced feed efficiency, reduced feed requirements per unit of weight gain, and mitigated methane emissions (Jayanegara et al., 2014). In contrast, the findings of the present study diverge from those of Lind et al. (2024), who reported no significant effect of biochar supplementation on in vitro and in vivo methane production or lamb growth rates across four different biochar levels (0, 10, 23, and 46 g/kg DM).

4. Conclusion

Developing effective food systems and ensuring food security have become global imperatives for all nations. This effort encompasses enhancing both crop and livestock production. However, expanding livestock production presents challenges, particularly concerning greenhouse gas emissions. The livestock sector alone contributes to 14.5% of total anthropogenic greenhouse gas emissions. In this study, the addition of biochar to animal rations has proven to be an effective solution, with daily supplementation of 1% biochar. The research demonstrated significant reductions in methane gas emissions and improvements in average daily weight gain. Biochar was found to enhance nutrient utilization, improve feed efficiency, and mitigate methane emissions in livestock. Consequently, incorporating biochar into sheep diets was observed to modify digestive processes and enhance nutrient absorption. This study highlights biochar's potential as a sustainable approach to mitigating environmental challenges associated with livestock production.

Table 5
Methane gas emissions from both groups of animal in (ppm).

Days	0	7	14	21	28	35	42	49	56	63
Control	1082± 0.341	1062± 0.367	1020± 0.674	1089± 0.120	1090± 0.324	1084± 0.837	1043± 0.205	1051± 0.398	1093± 0.189	1096± 0.781
CH_4 (ppm) + biochar	1082± 0.021	636± 0.329	616± 0.028	657± 0.298	611± 0.209	614± 0.213	624± 0.219	627± 0.393	642± 0.123	617± 0.987
% reduction compared to control group	–	66.98	65.58	65.75	78.39	76.55	67.15	67.62	70.25	77.63

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CRediT authorship contribution statement

Hana'a Burezq: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Fatin Khalil:** Data curation, Software.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dr. Hana'a A. Burezq reports financial support was provided by Kuwait Institute for Scientific Research. Dr. Hana'a A. Burezq reports equipment, drugs, or supplies was provided by the Public Authority of Agriculture Affairs and Fish Resources (PAAFR). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Abbas, S., Razzaque, M.A., Elsaid, A., Albert, S., Gelan, A., Khalil, F., Naseeb, A., Burezq, H., 2015. The effect of nutrition and season of breeding on reproductive performance of Naeemi ewes. Final Report. Kuwait Institute for Scientific Research. KISR No. 12848.

Al-Khalaf, H., Nasser, A., 2023. Critical review on the use of biochar in poultry industry: benefits, characteristics, and applications. World Poultry Sci. J. 79 (4), 807–833. https://doi.org/10.1080/00439339.2023.2239776.

Al-Kindi, A., Schiborra, A., Buerkert, A., Schlecht, E., 2016. Effects of quebracho tannin extract and activated charcoal on nutrient digestibility, digesta passage and faeces composition in goats. J. Anim. Physiol. Anim. Nutr. 101 (3), 576–588. https://pubmed.ncbi.nlm.nih.gov/26852897/.

Al-Naffakh, J., Al-Qassab, M.R., Al-Makhzoomi, Z.M.H., 2021. Share of CO_2 emissions in Iraq, Saudi Arabia and Kuwait from the combustion of fossil fuels: a statistical study. J. Res. Technol. Educ. 2 (3), 114–120.

Besharati, M., Palangi, V., Niazifar, M., Nemati, Z., 2020. Comparison study of flaxseed, innamon and lemon seed essential oils additives on quality and fermentation characteristics of lucerne silage. Acta. Agriculture Slovenica 115, 2. http://ojs.aas.bf.uni-lj.si/index.php/AAS/article/view/1483.

Connor, E.E., 2015. Invited review: improving feed efficiency in dairy production: challenges and possibilities. Animal 9 (3), 395–408. https://doi.org/10.1017/S1751731114002997.

Das, N.G., Sarker, N.R., Najmul Haque, N. Md, 2020. M. D. An estimation of greenhouse gas emission from livestock in Bangladesh. J. Adv. Vet. Anim. Res. 7 (1), 133–140. https://doi.org/10.5455/javar.2020.g402. PMCID: PMC7096121. PMID: 32219119.

Diatta, A.A., Fike, J.H., Battaglia, M.L., Galbraith, J.M., M, J., Baig, M.B., 2020. M. B. Effects of biochar on soil fertility and crop productivity in arid regions: a review. Arabian J. Geosci. 13, 595.

Downie, A., Munroe, P., Cowie, A., Van Zwieten, L., Lau, D.M.S., 2012. Biochar as a geoengineering climate solution: hazard identification and risk management. Crit. Rev. Environ. Sci. Technol. 42, 225–250. https://www.tandfonline.com/doi/abs/10.1080/10643389.2010.507980.

Elghandour, M.M.M.Y., Adegbeye, M.J., Barbabosa-Pilego, A., Perez, N.R., Hernández, S. R., Zaragoza-Bastida, A., Salem, A.Z.M., 2019. Equine contribution in methane emission and its mitigation strategies. J. Equine Vet. Sci. 72, 56–63. https://www.sciencedirect.com/science/article/pii/S0737080618306452.

Elkhalifa, S.H.E., Al-Ansari, T., Mackey, H.R., Mckay, G., 2019. Food waste to biochars through pyrolysis. A review 310–320. https://doi.org/10.1016/j.resconrec.2019.01.024.

Hansen, H.H., Storm, I.M.L.D., Sell, A.M., 2012. Effect of biochar on in vitro rumen methane production. Acta Agric. Scand. Sect. A Anim. Sci. 62, 453. https://www.tandfonline.com/doi/abs/10.1080/09064702.2013.789548.

Harbi, M., Bin Shams, M., Alhajri, I., 2020. Connectionist modelling for anthropogenic greenhouse gases (GHG) emissions in urban environments. ISSN 1589-1623. 10. Appl. Ecol. Environ. Res. 18 (2), 2087. https://doi.org/10.15666/aeer/1802_20872107.

Herbertson, N., 2018. Biochar as a ruminant feed supplement to improve animal health, soil quality, and net greenhouse gas emissions, 1, 1–20.

IPCC, 2013. Climate Change the Physical Science Basis. American Geophysical Union, Fall Meeting 2007 abstract id. U43D-01. CRID: 1381413280484207372. https://cir.nii.ac.jp/crid/1381413280484207372.

Jayanegara, A., Wina, E., Takahashi, J., 2014. Meta-analysis on methane mitigating properties of saponin-rich sources in the rumen: influence of addition levels and plant sources. Asian-australas. J. Anim. Sci. 27 (10), 1426–1435. https://doi.org/10.5713/ajas.2014.14086.

Joseph, S., Kammann, C.I., Shepherd, J.G., Conte, P., Schmidt, H.-P., Hagemann, N., Rich, A.M., Marjo, C.E., Allen, J., Munroe, P., Mitchell, D.R.G., Donne, S., Spokas, K., Graber, E.R., 2018. Microstructural and associated chemical changes during the composting of high-temperature biochar: mechanisms for nitrate, phosphate and other nutrient retention and release. Sci. Total Environ. 618, 1210–1223. https://pubmed.ncbi.nlm.nih.gov/29126641/.

Kammann, C.I., Schmidt, H.-P., Messerschmidt, N., Linsel, S., Steffens, D., Müller, C., Koyro, H.-W., Conte, P., Erratum, S. Joseph, 2015. Plant growth improvement mediated by nitrate capture in co-composted biochar. Sci. Rep. 5 (1), 12378. https://www.researchgate.net/publication/276992968_Erratum_Plant_growth_improvement_mediated_by_nitrate_capture_in_co-composted_biochar.

Keba, D., Tolemariam, T., Demeke, S., Alkhtib, A., 2023. A. Corncob biochar supplementation improves nutrient digestibility, fattening performance and carcass characteristics of fattening sheep. Veterinary Medicine and Science 9 (2), 967–973. https://doi.org/10.1002/vms3.1072. Epub 2023 Jan 9.

Lenga, R.A., Preston, T.R., Inthapanya, S., 2012a. Biochar reduces enteric methane and improves growth and feed conversion in local "Yellow" cattle fed cassava root chips and fresh cassava foliage. Livest. Res. Rural Dev. 24, 11. https://www.researchgate.net/publication/287508963_Biochar_reduces_enteric_methane_and_improves_growth_and_feed_conversion_in_local_Yellow_cattle_fed_cassa.

Lengb, R.A., Inthapanya, S., Preston, T.R., 2012b. Methane production is reduced in an *in vitro* incubation when the rumen fluid is taken from cattle that previously received biochar in their diet. Livest. Res. Rural Dev. 24, 11. https://www.lrrd.cipav.org.co/lrrd24/11/sang24211.htm.

Lind, V., Sizmaz, O., Demirtas, A., Sudagidan, M., Weldon, S., Budai, A., O'Toole, A., Miladinovic, D.D., Jørgensen, G.M., 2024. Biochar effect on sheep feed intake, growth rate and ruminant in vitro and in vivo methane production. Animal 18 (6), 101195, 2024.

Llonch, P., Haskell, M.J., Dewhurst, R.J., Turner, S.P., 2016. Current available strategies to mitigate greenhouse gas emissions in livestock systems: an animal welfare perspective. Cambridge University Press. Animal 11 (2), 274–284. Published online by.

Luna, I.C., Farma, F., Waterhouse, A., Sohi, S., Rooke, J., 2018. Effect of biochar produced from different biomass sources and at different process temperatures on methane production and ammonia concentrations in vitro. Anim. Feed Sci. Technol. 237. https://doi.org/10.1016/j.anifeedsci.2018.01.003.

Man, K.Y., Chow, K.L., Man, Y.B., Mo, W.Y., Wong, M.H., M, H., 2020. Use of biochar as feed supplements for animal farming. Crit. Rev. Environ. Sci. Technol. 51 (2), 187–217. https://doi.org/10.1080/10643389.2020.1721980, 2020.

McAvoy, D.J., Burritt, B., Villalba, J.J., 2020. Use of biochar by sheep: impacts on diet selection, digestibility, and performance. J. Anim. Sci. 98 (12), 1–9. https://academic.oup.com/jas/article/98/12/skaa380/5998260.

Mirheidari, A., Torbatinejad, N.M., Shakeri, P., Mokhtarpour, A., 2020. Effects of biochar produced from different biomass sources on digestibility, ruminal fermentation, microbial protein synthesis and growth performance of male lambs. Small Rumin. Res. 183, 106042. https://www.researchgate.net/publication/338058353_Effects_of_biochar_produced_from_different_biomass_sources_on_digestibility_ruminal_fermentation_microbial_pro.

Moss, A.R., Jouany, J.-P., Newbold, J., 2000. Methane production by ruminants: its contribution to global warming. Zootech 49, 231–253. https://animres.edpsciences.org/articles/animres/pdf/2000/03/z0305.pdf.

Nawaz, A., Li, E., Irshad, S., Xiong, Z., Xiong, H., Shahbaz, H.M., Siddique, F., 2020. Valorization of fisheries by-products: challenges and technical concerns to food

industry. Trends Food Sci. Technol. 99, 34–43. https://www.frontiersin.org/articles/10.3389/fsufs.2021.611835.

Osman, A.I., Fawzy, S., Farghali, M., El-Azazy, M., Elgarahy, A.M., Fahim, R.A., Abdel Maksoud, M.I.A., Ajlan, A.A., Yousry, M., Saleem, Y., Rooney, D.W., 2022. Biochar for agronomy, animal farming, anaerobic digestion, composting, water treatment, soil remediation, construction, energy storage, and carbon sequestration: a review. Environ. Chem. Lett. 20, 2385–2485.

O Toole, A., Andersson, D., Gerlach, A., Glaser, B., Kammann, C.I., Kern, J., Kuoppamäki, K., Mumme, J., Schulze, M.S.H.-P., Stenrød, M.S.F., Stenström, J., 2016. Current and future applications for biochar. In: Shackley, S., Ruysschaert, G., Zwart, K., Glasereds, B. (Eds.), Biochar in European Soils and Agriculture: Science and Practice. Taylor & Francis Group, Abington, pp. 253–280. https://www.researchgate.net/publication/295812792_Current_and_future_applications_for_biochar.

Parthasarathy, P., Alherbawi, M., Shahbaz, M., Al-Ansari, T., McKay, G., 2023. Developing biochar from potential wastes in Qatar and its revenue potential. Energy Convers. Manag. 20, 100467. https://doi.org/10.1016/j.ecmx.2023.100467.

Pereira, R.C., Muetzel, S., Arbestain, M.C., Bishop, P., Hina, K., Hedley, M., 2014. Assessment of the influence of biochar on rumen and silage fermentation: a laboratory-scale experiment. Anim. Feed Sci. Technol. 196, 22–31. https://www.sciencedirect.com/science/article/abs/pii/S0377840114002247.

Pinares-Patiño, C.S., Ebrahimi, S.H., McEwan, J.C., Clark, H., Luo, D., 2011. Is rumen retention time implicated in sheep differences in methane emission? Proc. N. Z. Soc. Anim. Prod. 71, 219–222. https://doi.org/10.13140/2.1.3309.5043.

Pinares-Patiño, C.S., Hunt, C., Martin, R., West, J., Lovejoy, P., Waghorn, G.C., 2012. New Zealand ruminant methane measurement center. In: Manual on Respiration Chambers Designs. CS Pinares- Patiño and GC Waghorn, pp. 9–28. GRA, New Zealand. Retrieved May 9, 2013, from. http://www.globalresearchalliance.org/app/uploads/2012/03/GRA-MAN-Facility-BestPract-2012-ch11.pdf.

Sakadevan, K., Nguyen, M.-L., 2017. Chapter four - livestock production and its impact on nutrient pollution and greenhouse gas emissions. Adv. Agron. 141, 147–184.

Schmidt, H.-.P., Hagemann, N., Draper, K., Kammann, C., 2019. The use of biochar in animal feeding. PMID: 31396445 PMCID: PMC6679646. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6679646/.

Silivong, P., Preston, T.R., 2016. Supplements of water spinach (Ipomoea aquatica) and biochar improved feed intake, digestibility, N retention and growth performance of goats fed foliage of Bauhinia acuminata as the basal diet. Livest. Res. Rural Dev. 28, 5. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9967708/.

Sivilai, B., Prestonl, T.R., 2019. Rice distillers' byproduct and biochar as additives to a forage-based diet for native Moo Lath sows during pregnancy and lactation. Livest. Res. Rural Dev. 31, 10. http://www.lrrd.org/lrrd31/10/sivil31151.html.

Tamayao, P.J., Ribeiro, G.O., McAllister, T.A., Yang, H.E., Saleem, A.M., Ominski, K.H., Okine, E.K., McGeough, E.J., 2021. Effects of post-pyrolysis treated biochar on methane production, ruminal fermentation, and rumen microbiota of a silage-based diet in an artificial rumen system (RUSITEC). Anim. Feed Sci. Technol. 273, 114802. https://www.sciencedirect.com/science/article/pii/S0377840120307070.

Tang, S.g, Ma, L., Wei, X., Tian, D., Wang, B., Li, Z., Zhang, Y., Shao, X., 2019. Methane emissions in grazing systems in grassland regions of China: a synthesis. Sci. Total Environ. 654, 662–670. https://doi.org/10.1016/j.scitotenv.2018.11.102.

Tapio, I., Snelling, T.J., Strozzi, F., Wallace, R.J., 2017. The ruminal microbiome is associated with methane emissions from ruminant livestock. J. Anim. Sci. Biotechnol. 8, 7. https://jasbsci.biomedcentral.com/articles/10.1186/s40104-017-0141-0.

Villalba, J.J., Provenza, F.D., Banner, R.E., 2002. Influence of macronutrients and activated charcoal on intake of sagebrush by sheep and goats. J. Anim. Sci. 80 (8), 2099–2109. https://pubmed.ncbi.nlm.nih.gov/12211378/.

Zhang, X., Gao, B., Zheng, Y., Hu, X., Creamer, A.E., Annable, M.D., Li, Y., 2017. Biochar for volatile organic compound (VOC) removal: sorption performance and governing mechanisms. Bioresour. Technol. 245, 606–614, 2017. https://www.researchgate.net/publication/319495940_Biochar_for_volatile_organic_compound_VOC_removal_Sorption_performance_and_governing_mechanisms.